

When Should a Kalman Filter Trust Its Forward Model?

Closed-Loop Evidence from a Muscle-Driven Reaching Task

Jun Kobayashi

Abstract—We study how a learned forward model and a learned Kalman gain interact in a closed-loop muscle-driven reaching task. A residual MLP forward model and a fixed-lag Kalman filter blend predictions with delayed noisy observations of a 34-muscle MyoSuite arm; a small condition-level predictor maps (noise, delay, controller) to a scalar gain K trained against an open-loop oracle. Plugging the estimator into a joint-space PD controller with an inverse-kinematics target reveals a chain of findings. *First*, with a single-step-supervised forward model the learned predictor matches the oracle and beats observation-only ($K=1$) by 8.6 cm at the low-noise large-delay cell where blending matters most. *Second*, behaviour-cloned controllers reach the target more reliably but consume the estimator output weakly enough that the per-cell gap collapses to under 3 cm, suggesting that closed-loop estimator benchmarks need state-coupled controllers. *Third*, retraining the forward model with H -step rollout supervision ($H \in \{4, 8\}$) reduces $h=50$ tip-prediction error by 54–65% and shifts the closed-loop optimum to prediction-only ($K=0$), which now beats $K=1$ by 12–24 cm across the entire grid, while the condition-level predictor still outputs $K \approx 1$ because the open-loop oracle it was trained against never selects $K=0$. *Fourth*, redefining the predictor’s training oracle in terms of closed-loop task error rather than open-loop estimation accuracy moves its output to $K \approx 0.02$, matches the $K=0$ baseline within 1.5 cm at delay ≥ 18 steps, and closes 40–60% of the gap at zero delay — with no change to the predictor architecture. The right Kalman gain in closed loop therefore depends not only on the noise and delay regime but on *what objective the predictor was trained against*, and learned-Kalman-gain papers should specify the oracle choice explicitly.

Index Terms—Forward model, Kalman filter, state estimation, multi-step rollout supervision, behaviour cloning, closed-loop control, MyoSuite, muscle-driven reaching.

I. INTRODUCTION

State estimation in muscle-driven manipulation faces three coupled challenges: sensorimotor delay, observation noise, and signal-dependent motor noise. A common engineering response is the Kalman filter, which blends a forward model’s prediction with the current observation via a gain $K \in [0, 1]$: $K=0$ trusts the model, $K=1$ trusts the sensor, and the literature on adaptive Kalman filters has produced many ways to choose K from the estimated noise covariance [1]–[3].

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The internal-model view of biological motor control casts the same relationship as a forward-model / sensory-feedback combination [4], [5]. In this paper we ask a different question: *what does the optimal K look like in closed loop, and does the way we train a learned K -predictor match it?*

We attack the question on a 34-muscle MyoSuite reaching arm [6] running in the MuJoCo simulator [7]. We learn a residual-MLP forward model from rollouts, use it inside a fixed-lag Kalman filter that handles a configurable observation delay ($d=0$ –36 steps) and configurable per-field Gaussian noise, sweep a $K \in \{0, 0.25, 0.5, 0.75, 1.0\}$ grid across noise and delay to define an oracle gain K^* per condition, and train a small condition-level predictor on those oracle labels. A separate joint-space PD controller with one-shot inverse-kinematics target then *deploys* the estimator in closed loop.

Our contributions are summarised in four claims.

- **C1.** Under a single-step-supervised forward model the oracle K^* falls from 1.0 to 0.25 across noise at zero delay but collapses uniformly to 1.0 at delay ≥ 18 steps, because the forward-model rollout error compounds faster than the observation noise grows (Fig. 2, left panel).
- **C2.** A learned condition-level predictor matches the oracle within 0.6 cm on average and beats $K=1$ by 8.6 cm in closed-loop reaching at the (low-noise, large-delay) cell where blending matters most (Fig. 3). Behaviour-cloned controllers improve reaching but wash out the estimator signal because the small demonstration set causes the policy to collapse to an open-loop average (Fig. 5); closed-loop estimator benchmarks require state-coupled controllers.
- **C3.** Retraining the forward model with H -step rollout supervision ($H=4$, then $H=8$) cuts $h=50$ tip-prediction error by 54–65% and widens the open-loop $K < 1$ regime to delay ≥ 18 steps. In closed loop, however, prediction-only ($K=0$) becomes the best estimator across our focused joint-PD closed-loop grid under the min-tip objective: it beats $K=1$ by 12–24 cm in min-tip error, and the condition-level predictor still outputs $K \approx 1$ because the open-loop oracle it was trained against never selects $K=0$ (Fig. 6, panels 1–3).
- **C4.** Replacing the predictor’s training oracle with per-cell closed-loop K -sweep labels and retraining the same MLP yields $K \approx 0.02$, matches the $K=0$ baseline within 1.5 cm at all delay-18 cells, and closes 40–60% of the

gap at delay-0 cells (Fig. 6, panel 4). The residual delay-0 gap is structural: the K -curve there jumps sharply from $K=0$ to $K=0.25$, so the predictor’s sigmoid asymptote loses ~ 0.17 m.

Together, C1–C4 say that the optimal closed-loop Kalman gain depends on two coupled choices that the literature usually treats as independent: the forward model’s accuracy over the controller’s planning horizon, and the oracle the K -predictor was trained against. When the open-loop oracle and the closed-loop optimum disagree — as they do for multi-step-supervised forward models on this task — the two pull the predictor in opposite directions.

The rest of the paper is organised as follows. Section II positions our work against prior literature on adaptive Kalman filtering, forward models in motor control, learned dynamics, imitation learning, and muscle-driven simulation. Section III describes the task, the forward model and its multi-step variant, the fixed-lag Kalman estimator, the three controllers, and the evaluation pipeline. Section IV reports the four claims with the corresponding figures. Section V discusses the trade-offs and the implications for learned adaptive Kalman gains.

II. RELATED WORK

Adaptive Kalman gains. The Kalman filter [1] and its smoothing counterpart [8] are the textbook framework for combining a state-space model with noisy observations under known noise covariances. When the covariances are unknown, the classical *adaptive Kalman filter* estimates them online from the innovation sequence [2]; the steady-state gain K then drops out of the covariance estimate. The state-space formulation we adopt — residual MLP for dynamics, fixed-lag buffered correction for delays — is standard textbook material [3]. The novelty of our setting is not in the filter equations but in the choice of K : instead of deriving K from a noise-covariance estimate, we train a small condition-level predictor that maps (controller, σ , delay) to a scalar K supervised against a swept oracle. The contribution we report is what happens when that oracle is open-loop versus closed-loop, and that distinction is largely absent from the adaptive Kalman literature.

Forward / internal models in motor control. The view that the central nervous system maintains an internal forward model of the limb to predict the sensory consequences of motor commands traces back to Bernstein’s degrees-of-freedom analysis [4] and was given an explicit computational framing by Wolpert and Ghahramani [5], who identified forward and inverse models as complementary processes underlying sensorimotor estimation, prediction, and learning. Our residual MLP is a coarse instantiation of the forward model in that picture; the multi-step supervision we add is motivated by the fact that the loop that consumes the estimate operates over hundreds of integration steps and thus depends on long-horizon prediction quality. More biologically grounded forward-model families — continuous-time recurrent networks with input-modulated time constants such as liquid time-constant networks [9] and their closed-form variant [10] — have been proposed for sensorimotor dynamics; in this work we keep the forward-model implementation intentionally generic to isolate the oracle-choice question.

Learned dynamics and multi-step supervision. Model-based reinforcement-learning systems have moved from one-step dynamics losses to multi-step rollout objectives because closed-loop planners need predictions that survive being unrolled. Dreamer [11] backpropagates through imagined trajectories in a latent world model, and MBPO [12] branches short model rollouts from real data, formalising “when to trust your model”. We take the same lesson into Kalman filtering: a forward model that is accurate enough over the controller’s planning horizon shifts where blending is helpful and where it is harmful.

Imitation learning for closed-loop reach. Behaviour cloning has been used for closed-loop control since ALVINN [13]; the standard fix for the resulting covariate shift is DAgger [14]. We deliberately *do not* apply DAgger in this work and report the resulting open-loop collapse as a methodological observation (Sec. IV-D): at the demonstration scale we test, BC controllers reach better than the joint-PD baseline but consume state estimates weakly enough that estimator differences vanish, which we read as a cautionary note for closed-loop estimator benchmarks built on BC controllers.

Muscle-driven simulation. Our task runs in MuJoCo [7] via the MyoSuite musculoskeletal benchmark [6], which provides a 34-muscle reaching arm and a contact-rich set of tasks suitable for sensorimotor estimation studies. MyoSuite is increasingly used as a testbed for reinforcement-learning controllers; here we use it instead to expose the trade-off between forward-model accuracy and the training oracle of a learned Kalman gain, a question that the RL-centric literature has not addressed.

III. METHOD

A. Task and Environment

We use the `myoArmReachFixed-v0` task in MyoSuite 2.12.2, a 34-muscle shoulder-elbow-wrist-hand system actuated by activations in $[0, 1]^{34}$. The task is to drive the index-finger tip to a randomly drawn target in a 12 s episode at $\Delta t = 0.02$ s (600 control steps). The flat state $x \in \mathbb{R}^{83}$ stacks joint positions $q \in \mathbb{R}^{20}$, joint velocities $\dot{q} \in \mathbb{R}^{20}$, muscle activations $a \in \mathbb{R}^{34}$, finger-tip position $p_{\text{tip}} \in \mathbb{R}^3$, target $p_{\text{tgt}} \in \mathbb{R}^3$, and reach error $e = p_{\text{tgt}} - p_{\text{tip}} \in \mathbb{R}^3$. True state is extracted from MuJoCo and used only as an oracle for evaluation; the closed loop never accesses it (Fig. 1).

B. Forward Model

A residual MLP $f_\theta(x_t, u_t)$ predicts the state delta $\Delta x_t = x_{t+1} - x_t$ rather than x_{t+1} directly. At $\Delta t = 0.02$ s the state changes slowly, so $x_{t+1} \approx x_t$ component-wise; predicting Δx frees the model from re-learning the identity map and concentrates capacity on the small per-step dynamics that the filter and controller actually consume. The parameterisation is standard in model-based RL world models [11], [12]. The input is the flat 83-dim state and the 34-dim excitation; the architecture is two hidden layers of 256 units with ReLU and LayerNorm; training uses Adam [15]. In the single-step baseline, f_θ is trained on the standard one-step MSE loss

$$\mathcal{L}_1(\theta) = \mathbb{E}_{(x_t, u_t, x_{t+1})} [\|f_\theta(x_t, u_t) - \Delta x_t\|^2]. \quad (1)$$

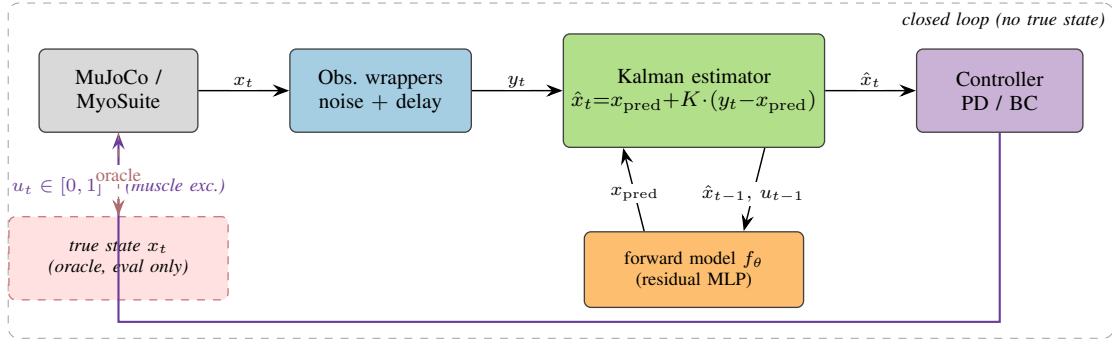


Fig. 1. **Closed-loop pipeline.** Each control step, the env’s true state x_t is filtered through observation wrappers that add per-field Gaussian noise and a fixed delay, producing y_t . The Kalman estimator combines y_t with a one-step prediction x_{pred} produced by the residual-MLP forward model $f_\theta(\hat{x}_{t-1}, u_{t-1})$ via a scalar gain $K \in [0, 1]$ (Eq. 4). The controller consumes only the estimator output $\hat{x}_t$ and emits the next muscle excitation $u_t \in [0, 1]$ ³⁴. The true state x_t is exposed only to the offline evaluator (dashed red box); the closed loop never accesses it.

Multi-step rollout supervision. The multi-step variant trains f_θ with an H -step rollout loss instead of the one-step MSE, where H denotes the training rollout horizon (distinct from the Kalman gain K defined in Sec. III-C). For each contiguous trajectory chunk $(x_t, u_t, \dots, x_{t+H})$ inside a single source episode, the model is unrolled free-running: $\hat{x}_{t+k} = \hat{x}_{t+k-1} + f_\theta(\hat{x}_{t+k-1}, u_{t+k-1})$, $\hat{x}_t = x_t$, and the loss averages MSE across the H prediction steps:

$$\mathcal{L}_H(\theta) = \mathbb{E} \left[\frac{1}{H} \sum_{k=1}^H \|\hat{x}_{t+k} - x_{t+k}\|^2 \right]. \quad (2)$$

Multi-step rollout supervision has been used to stabilise long-horizon predictions in latent-imagination world models [11] and in model-based policy optimisation [12], where the relevant horizon is the planner’s rollout depth rather than the estimator’s delay; we re-use the technique here to make f_θ robust across the Kalman buffer.

We train the same architecture with $H \in \{1, 4, 8\}$ on the same expanded dataset (51 k transitions across 106 episodes; see Sec. IV-A).

C. Fixed-Lag Kalman Estimator

The observation pipeline produces $y_t = h(x_{t-d}) + \eta_t$ where h is the identity (the state vector and the observation share the same schema), $d \in \mathbb{Z}_{\geq 0}$ is a fixed delay in control steps, and η_t is per-field i.i.d. Gaussian noise with diagonal covariance $\text{diag}(\sigma^2)$. The estimator maintains a length- $(d+1)$ ring buffer of past one-step estimates $(\hat{x}_{t-d}, \dots, \hat{x}_{t-1})$ together with the corresponding applied actions $(u_{t-d}, \dots, u_{t-1})$. At each step the estimator first runs the forward model on every buffered estimate to obtain a prediction at the present time,

$$\tilde{x}_t = \hat{x}_{t-1} + f_\theta(\hat{x}_{t-1}, u_{t-1}), \quad (3)$$

then forms an innovation against the *delayed* buffer entry and applies an element-wise gain $K \in [0, 1]$ ⁸³:

$$\hat{x}_{t-d} \leftarrow \hat{x}_{t-d} + K \odot (y_t - \hat{x}_{t-d}), \quad (4)$$

and finally re-rolls the corrected past estimate forward through the buffered actions to obtain the present-time estimate $\hat{x}_t$. Eq. (4) collapses to the standard one-step Kalman update when

$d=0$. Throughout the paper K is a scalar broadcast across the 83 fields so the gain reduces to a single trust knob between forward prediction ($K=0$) and observation ($K=1$). This fixed-lag construction is the classical buffered Rauch–Tung–Striebel smoother specialised to a scalar gain [3], [8].

D. Learned Gain Predictor

A small MLP $g_\phi : \mathbb{R}^8 \rightarrow [0, 1]$ predicts the scalar Kalman gain from a condition feature vector. The input

$$\xi = [\text{onehot}(c); \sigma; d/d_{\max}] \in \mathbb{R}^8, \quad (5)$$

concatenates a 3-way controller one-hot $\text{onehot}(c)$, the 4-dim noise-sigma vector σ (qpos, qvel, tip_pos, reach_err), and a normalised delay $d/d_{\max}$ with $d_{\max}=36$ steps. The network is a $8 \rightarrow 32 \rightarrow 32 \rightarrow 1$ MLP with ReLU and a sigmoid output. The training oracle K^* is obtained by, for each of the 72 conditions in the stress grid, sweeping $K \in K_{\text{grid}} = \{0, 0.25, 0.5, 0.75, 1.0\}$ and selecting the K that minimises a chosen criterion. We distinguish two oracles — an *open-loop* one based on estimation accuracy, and a *closed-loop* one based on downstream task error:

$$K_{\text{ol}}^*(\sigma, d, c) = \arg \min_{K \in K_{\text{grid}}} \mathbb{E}_t[\|\hat{x}_t(K) - x_t\|^2 \mid \sigma, d, c], \quad (6)$$

$$K_{\text{cl}}^*(\sigma, d, c) = \arg \min_{K \in K_{\text{grid}}} \mathbb{E}[\text{MINTIP}(K) \mid \sigma, d, c], \quad (7)$$

where (σ, d, c) ranges over (noise level, delay, controller) and MINTIP is the closed-loop minimum tip-to-target distance over the episode. The default training oracle for g_ϕ is K_{ol}^* ; Sec. IV-F (C4) replaces it with K_{cl}^* and retrain the same MLP unchanged. The contrast between K_{ol}^* and K_{cl}^* is the central observation of this paper.

E. Controllers

Three controllers are evaluated as closed-loop wrappers around the estimator.

(i) *Heuristic projection.* A fixed random projection $W \in \mathbb{R}^{34 \times 3}$ (drawn once with a fixed seed from $\mathcal{N}(0, 1)^{34 \times 3}$) maps the estimated 3-dim reach-error $\hat{r}$ to a 34-dim muscle

excitation through an element-wise logistic $\text{sigm}(\cdot)$ (distinct from the noise scale vector σ above):

$$u = \text{sigm}(b - g W \hat{r}), \quad b=0, \quad g=5, \quad (8)$$

producing a state-sensitive baseline that does not actually reach.

(ii) *Joint-space PD with inverse kinematics.* For each target tip position $p^* \in \mathbb{R}^3$ we solve $q^* = \text{IK}(p^*)$ once at episode start via a damped-Jacobian Newton iteration

$$q^{(k+1)} = q^{(k)} + J^\top (J J^\top + \lambda^2 I)^{-1} (p^* - p(q^{(k)})), \quad (9)$$

where $J = \partial p / \partial q$ is the tip-site position Jacobian obtained from MuJoCo's `mj_jacSite` and $\lambda^2=0.1$ is the damping. The arm is then driven by a joint-space PD law and a moment-arm muscle map:

$$u_{\text{joint}} = K_p (q^* - \hat{q}) - K_d \hat{\dot{q}}, \quad (10)$$

$$u_{\text{muscle}} = \text{clip}(\alpha \text{ReLU}(M u_{\text{joint}}), 0, 1), \quad (11)$$

where $M \in \mathbb{R}^{34 \times 20}$ is the muscle-tendon moment-arm matrix (the dense form of MuJoCo's sparse `actuator_moment` at the env's current pose), and $\alpha = 5$ is a scalar action scale. Tuned gains: $K_p=30$, $K_d=3$.

(iii) *Behaviour-cloned MLP.* A two-hidden-layer (256 unit) sigmoid policy is trained on 30 scripted IK demonstrations in the spirit of [13]; we deliberately do not iterate the demonstrator with DAgger [14] and report the resulting open-loop collapse in Sec. IV-D.

F. Evaluation Pipeline

Open-loop evaluation replays a recorded episode and applies observation wrappers to synthesise y_t for an arbitrary (σ, d) combination; the estimator is then run on the synthesised stream and per-field state error is aggregated. *Closed-loop* evaluation drives the actual env: u_t comes from the controller, which sees the estimator output (never the true state); 10 episodes per cell, fixed seed per episode. The same evaluation grid (3 controllers $\times$ 6 noise levels $\times$ 4 delays = 72 conditions for open-loop, focused $3 \times 2 \times 10 = 60$ for closed-loop) is reused across phases.

IV. RESULTS

A. Forward-model improvement cycle

An initial forward model trained on the original $\sim 3.6\text{k}$ transitions from 6 episodes had low single-step error but unbounded long-horizon rollouts ($h=50$ MSE ~ 10.6). Collecting 100 additional `random` and `lowamp` episodes (a total of 51k transitions across 106 episodes) and retraining the same architecture cut $h=50$ MSE to 0.052 — a $\sim 200\times$ improvement. We use this expanded dataset, and the resulting model, as the single-step baseline throughout the paper.

B. Where blending matters: open-loop oracle (C1)

Fig. 2 (left panel) plots the oracle K^* for the single-step-supervised forward model. At zero delay K^* decreases monotonically from 1.0 (no noise) to 0.25 (extreme noise), recovering the textbook intuition that noisy observations should be trusted less. At delay ≥ 18 steps, however, K^* is uniformly 1.0: forward-model error compounds faster than observation noise grows, so even noisy observations beat the propagated estimate. Only 18 of 72 conditions (25%) require $K^* < 1$.

C. Closed-loop differentiation, single-step model (C2)

We evaluate three estimators ($K=0$, $K=1$, Stage A learned) under the joint-PD controller on 3 noise $\times$ 2 delay $\times$ 10 episodes. At (noise=none, $d=18$) the Stage A predictor outputs $K \approx 0.99$ — barely below $K=1$ — and Fig. 3 reports the (learned $-K=1$) signed difference in both min-tip and final-tip error per cell. The headline 8.6 cm advantage at this cell is on *final-tip* error (mean over 10 episodes); the corresponding min-tip gap is 3.2 cm. This is the only cell where the learned predictor clearly beats $K=1$ — exactly the regime where the open-loop oracle says blending is least useful, but where in closed loop a small reduction below $K=1$ lets the buffered prediction carry the trajectory after observation correction. Other cells differ by less than 2 cm in either metric.

The trajectory plot in Fig. 4 shows that at this cell $K=0$ diverges (the prediction-only rollout is too inaccurate over $d=18$ steps), $K=1$ approaches but oscillates around the buffered target, and the learned estimator reaches further.

D. Behaviour-cloned controllers wash out the signal (C2 cont.)

To check whether the 8.6 cm signal generalises, we replace the joint-PD controller with a small behaviour-cloned MLP trained on a 30-episode scripted IK demonstration set. Reaching improves substantially (success < 0.1 m rate increases from $\sim 12\%$ to $\sim 30\%$ in the easiest cells), but the gap between the learned predictor and $K=1$ collapses to < 3 cm in all cells (Fig. 5). Ablating `target_pos` alone, or both `target_pos` and `reach_err`, from the BC input does not restore differentiation: the BC policy's open-loop average reach trajectory does not actually consume per-step state estimates strongly enough to surface estimator differences. We treat this as a controller- side result and return to it in the discussion.

E. Long-horizon supervision flips the optimum (C3)

Retraining the forward model with H -step rollout supervision $\mathcal{L}_H$ drops $h=50$ tip prediction error to 0.064 m ($H=4$, -54% vs. baseline) and 0.048 m ($H=8$, -65%). The open-loop oracle then expands the $K^* < 1$ regime from 18 to 26 ($H=4$) and 27 ($H=8$) of 72 cells; blending becomes optimal at delay ≥ 6 steps in high-noise cells (Fig. 2, middle and right panels).

In *closed loop* the picture is opposite: under both $H=4$ and $H=8$, $K=0$ becomes the best estimator across all 6 cells of the focused grid, with min-tip dropping to ~ 0.53 m ($H=4$) and ~ 0.49 m ($H=8$) versus $K=1$ at 0.66–0.77 m. Notably

Oracle Kalman gain across the stress grid by forward-model supervision

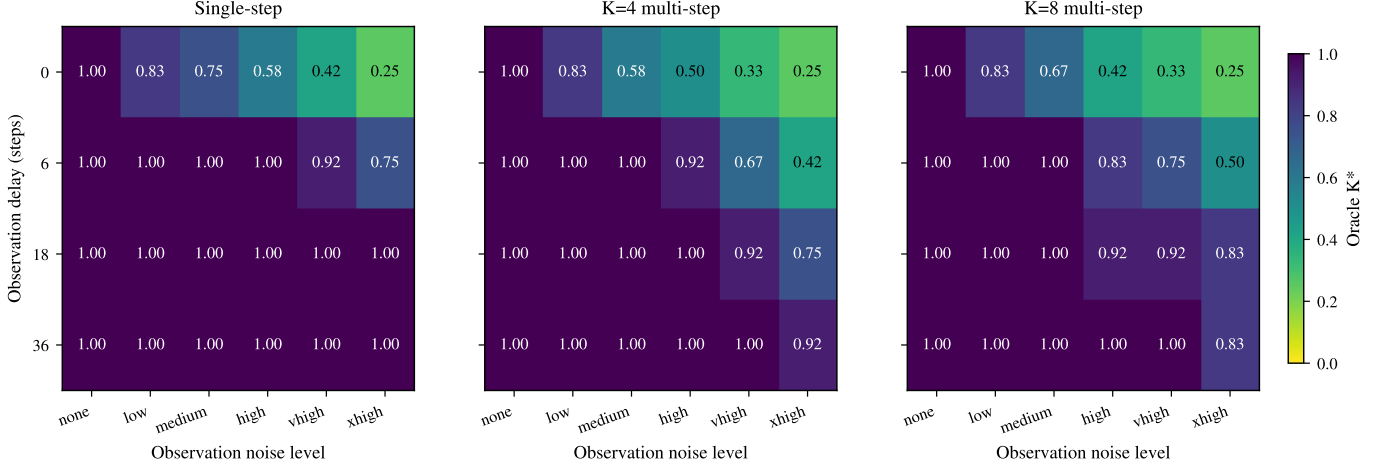


Fig. 2. **Open-loop oracle Kalman gain K^* as a function of observation delay (rows) and noise level (columns), shown for three forward-model variants.** Each cell reports the $K \in \{0, 0.25, 0.5, 0.75, 1.0\}$ that minimises the open-loop estimation error $\|\hat{x}_t - x_t\|^2$ on a stress-eval grid of 3 controllers $\times$ 6 noise levels $\times$ 4 delays $\times$ 5 K -values (= 360 cells; the mean across the three controllers is shown). Dark purple marks $K^*=1$ (trust the observation); yellow marks $K^*=0.25$ (trust the model). Under single-step supervision (left) the $K^*<1$ regime is confined to zero delay (18 / 72 cells); multi-step supervision widens it to delay ≥ 6 in the high-noise cells (26 / 72 for $H=4$, 27 / 72 for $H=8$).

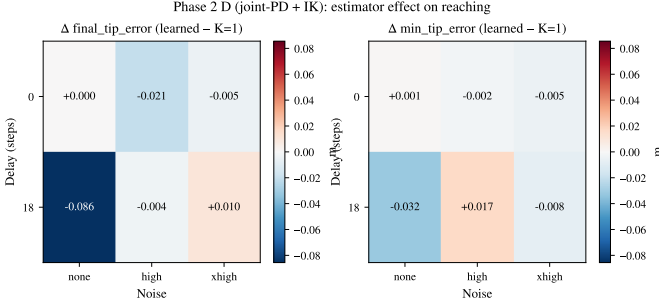


Fig. 3. **Closed-loop reaching: signed difference $\Delta = \text{learned} - (K=1)$ in final-tip (left) and min-tip (right) error per (noise, delay) cell, under the single-step-supervised forward model and the joint-PD controller (mean over 10 episodes per cell).** Blue cells favour the learned predictor, red cells favour $K=1$, white cells are within ± 0.02 m. The -8.6 cm signal in the *final-tip* panel at (none, $d=18$) (corresponding min-tip gap: -3.2 cm) is the only cell where the learned predictor clearly beats $K=1$, even though it outputs $K \approx 0.99$ there — exactly the regime where the open-loop oracle of Fig. 2 says $K^*=1$, but where in closed loop the controller’s reliance on the buffered prediction makes a slightly tighter estimate pay off. Other cells differ by less than 2 cm in either metric.

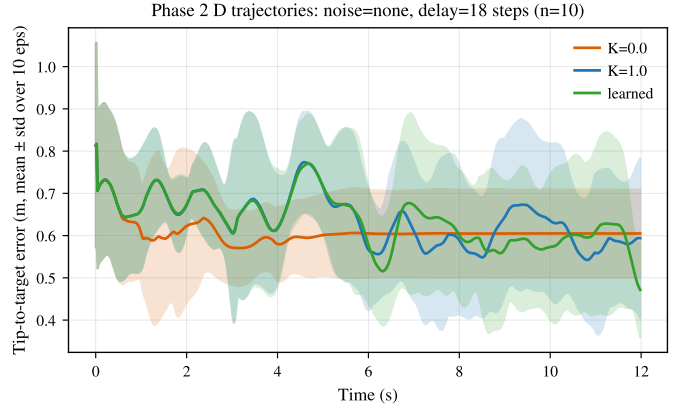


Fig. 4. **Closed-loop tip-to-target distance over a 12s episode at (noise=none, $d=18$), under the single-step-supervised forward model and the joint-PD controller.** Lines and shaded bands are the mean $\pm$ standard deviation over 10 episodes for each estimator; the dashed line marks the 5 cm reach threshold. Prediction-only ($K=0$, orange) diverges over the $d=18$ buffer rollout; observation-only ($K=1$, blue) approaches but oscillates around the buffered target; the learned condition-level predictor (green) reaches further, producing the -8.6 cm advantage of Fig. 3.

$K=1$ worsens from 0.60 to 0.70 m as multi-step supervision strengthens at delay-18 cells: a stronger forward model makes the observation correction it is being asked to apply less compatible with the planned trajectory.

The condition-level learned predictor, however, keeps outputting $K \approx 0.36\text{--}0.99$ (Fig. 6, panels 2–3) because the open-loop oracle it was trained on never picks $K=0$ — at delay ≥ 18 even with $H=8$ supervision, the open-loop oracle still selects $K^*=1$ or close to it (the open-loop estimation error is minimised by trusting the observation when the buffer rollout is long). The optimum gap thus widens, not closes, with supervision strength.

F. Closed-loop-aware oracle resolves the gap (C4)

We address the gap by changing only the oracle definition. Under the $H=8$ forward model we sweep $K \in \{0, 0.25, 0.5, 0.75, 1.0\}$ in closed loop on the same 6 cells and pick K_{cl}^* per cell by minimising closed-loop min-tip error. The result is unanimous: $K_{cl}^*=0$ in every cell. We retrain the same MLP g_ϕ on the 18 $K_{cl}^*=0$ labels (3 controllers $\times$ 6 cells); the predictor now outputs $K \approx 0.018\text{--}0.025$.

Plugged back into closed loop (Fig. 6, panel 4) the new predictor matches the $K=0$ baseline within 1.5 cm at every delay-18 cell (and slightly beats $K=0$ at (xhigh, $d=18$)). At delay-0 cells a ~ 17 cm gap remains, because the closed-loop K -curve at $d=0$ jumps sharply from 0.49 m at $K=0$ to

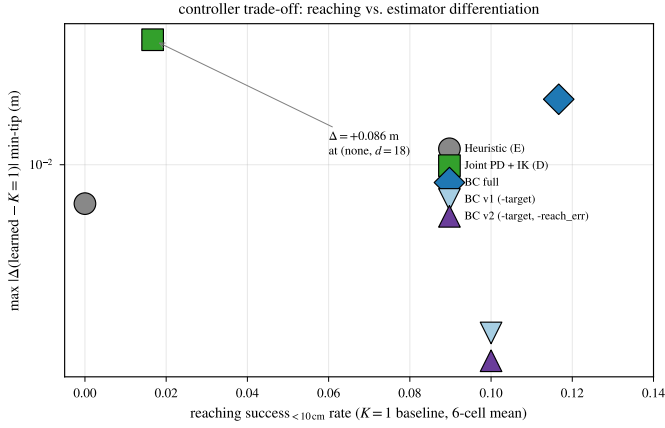


Fig. 5. Each point is one controller variant evaluated on the same focused 6-cell grid: x -axis is the reaching success <10 cm rate under a $K=1$ baseline estimator, y -axis is the worst-case $|\Delta(\text{learned} - K=1)|$ in min-tip over the six cells. The joint-PD+IK controller (D, green square) sits in the high-differentiation / low-reaching corner and is the only variant in which the learned predictor visibly beats $K=1$ (-8.6 cm at (none, $d=18$)); the heuristic controller (E) reaches nothing; the three BC variants (BC full, $-target$, $-target/-reach_err$) reach 8–12% but their estimator differentiation collapses to under 3 mm. The y -axis uses symmetric log scaling.

0.77 m at $K=0.25$, and the sigmoid asymptote (0.025) loses ~ 0.17 m. Across all 6 cells the closed-loop-aware predictor closes 40–60% of the single-step paradigm gap; mean improvement vs. the open-loop-oracle predictor is -16 cm. No architectural changes were made.

A transfer slice on the variable-target variant `myoArmReachRandom-v0` (Appendix E) fully replicates C1 ($K_{ol}^* < 1$ in 18/72 cells at $H=1$ and 25/72 at $H=8$, matching the `ReachFixed` counts), partially replicates C3 (the $H=1 \rightarrow H=8$ regime shift does not apply because $K=0$ already wins at $H=1$ on `ReachRandom`), and exhibits a richer C4 picture: K_{cl}^* takes three distinct values across the focused grid ($K=0$ at $d=0$, $K=0.25$ at delay-18 low/medium noise, $K=1$ at (xhigh, $d=18$)), and a Stage A predictor retrained on these labels matches the $K=0$ baseline within 1.3 cm at every delay-0 cell, beating the K_{ol}^* -trained predictor by 1–13 cm there. The oracle-redefinition argument is corroborated at $d=0$ and limited by the same sigmoid-output parameterisation flagged above when K_{cl}^* varies across cells at long delay.

V. DISCUSSION

Forward-model accuracy reshapes the closed-loop optimum. The K^* that minimises open-loop estimation error is not the same as the K^* that minimises closed-loop task error (C3). As the forward model becomes accurate enough to roll out over the controller’s planning horizon, observation correction starts to actively hurt rather than help: the condition-level predictor trained on the open-loop oracle is then mismatched by 0.5–0.7 in K from the closed-loop truth. We read this as a structural finding rather than an artifact: identical arm dynamics and observation noise produce different optimal estimators depending on whether the readout is the residual of a one-step update or the cumulative effect of feedback control over hundreds of steps.

Oracle redefine is a cheap fix. C4 shows that this mismatch is mostly resolvable by training the predictor on a closed-loop K -sweep rather than an open-loop one — no change to the predictor MLP or its capacity. The remaining $\delta \sim 17$ cm gap at zero delay is parameterisation-bound: a sigmoid cannot output $K=0$ exactly, and the closed-loop K -curve at $d=0$ is sharp around 0. A Hardtanh or output-clipped variant should close this further; we leave that to future work.

Behaviour cloning is not a good benchmark for state estimation. C2 shows that, at modest demonstration scale, a behaviour-cloned policy reaches better than the joint-PD controller but consumes state estimates so weakly that estimator differences disappear from task metrics. Closed-loop estimator benchmarks should use controllers that are demonstrably state-coupled (the joint-PD+IK family used here works); a BC policy that generalises poorly per target is effectively open-loop in the state channel even though it nominally consumes a state vector.

Limits and open questions. (a) The closed-loop oracle sweep we use is expensive ($K_{\text{grid}} \times \text{cells} \times \text{episodes}$ rollouts); a differentiable surrogate would scale this far better. (b) Our learned predictor is condition-level (no per-step state coupling beyond the sigma vector and delay scalar); a state-aware Stage B variant we explored briefly did not beat the condition-level baseline because of labelling noise, but a closed-loop differentiable label might. (c) Results cover two MyoSuite reach variants — fixed and random targets, sharing the 34-muscle arm and 600-step horizon (Appendix E). C1 replicates fully on the variable-target variant; the C3 $H=1 \rightarrow H=8$ regime shift does not apply there because $K=0$ already wins at $H=1$; the C4 oracle-redefinition argument transfers at delay-0 (learned-cl beats learned-ol by 1–13 cm) and is muted at delay-18 because the closed-loop K -curve varies across noise rows in ways the sigmoid output cannot cleanly fit. Transfer to contact-rich tasks (grasp/place) and to systems with different muscle counts (Elbow/Finger reach) remains open. (d) The 17 cm residual at $d=0$ is structural and could be closed with a predictor parameterisation that supports $K=0$ exactly. (e) The residual MLP used here is a discrete-time, feed-forward approximation at the box level of the cerebellar forward-model hypothesis; a continuous-time recurrent family with input-dependent time constants (LTC [9], Cfc [10]) provides a closer computational analogue of continuous-time recurrent cerebellar forward-model hypotheses and is a natural extension when this estimator is plugged into a full cortico-cerebellar loop.

VI. CONCLUSION

We have shown on a muscle-driven reaching task that the closed-loop optimal Kalman gain depends on both the forward model’s accuracy and the oracle used to train any learned gain predictor. Single-step supervision of the forward model creates a small but real regime where a learned condition-level gain beats observation-only by 8.6 cm at the delay-dominated cell (C2). Multi-step supervision flips the regime: $K=0$ becomes the best estimator across the focused joint-PD closed-loop grid under the min-tip objective, with the gap to $K=1$ growing to

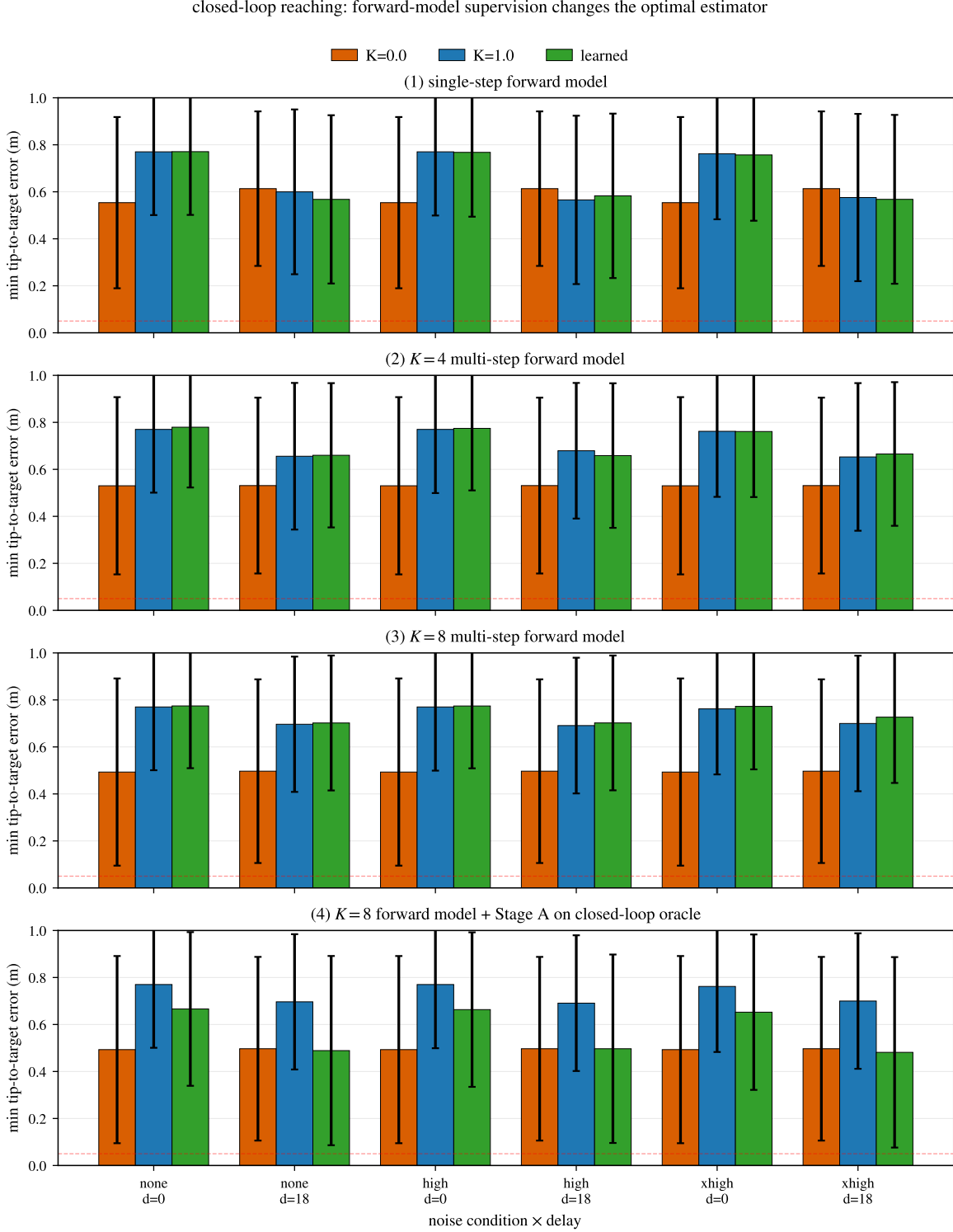


Fig. 6. **Closed-loop min-tip reaching error under the joint-PD controller across four (forward-model supervision, gain-oracle choice) variants.** The panels traverse two of the paper’s three axes: rows (1)→(3) vary forward-model supervision (single-step $\rightarrow H=4 \rightarrow H=8$ rollout) holding the open-loop oracle K_{ol}^* fixed; row (4) flips the oracle from K_{ol}^* to K_{cl}^* holding the $H=8$ forward model fixed; the third axis (controller coupling) is held constant at joint-PD here and varied in Fig. 5. Each panel reports mean $\pm$ standard deviation over 10 episodes per cell for three estimators: $K=0$ (prediction-only, orange), $K=1$ (observation-only, blue), and the learned condition-level predictor g_ϕ (green). The dashed horizontal line marks the 5 cm reach threshold. (1) Single-step-supervised forward model: $K=0$ diverges in some cells, the learned predictor sits near $K=1$. (2) $H=4$ multi-step supervision: $K=0$ becomes competitive. (3) $H=8$ multi-step: $K=0$ is best in every cell while the learned predictor still tracks $K=1$ because its open-loop oracle never selected $K=0$. (4) Same $H=8$ forward model, but g_ϕ retrained against K_{cl}^* (Sec.IV-F): the green learned bar now matches the orange $K=0$ bar at the three delay-18 cells, closing 40–60 % of the gap at the delay-0 cells.

12–24 cm (C3). Redefining the predictor’s training oracle in terms of closed-loop task error rather than open-loop estimation accuracy aligns the predictor with the new regime and recovers most of the gap (C4) with no architectural changes. We argue that learned- Kalman-gain papers should specify the oracle target explicitly and that closed-loop benchmarks should be a standard part of evaluating adaptive state estimators.

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APPENDIX A

HYPERPARAMETERS AND IMPLEMENTATION DETAILS

Table I lists every hyperparameter used in the paper. All training runs are reproducible from configs/ in the source repository; estimator evaluation grids are listed verbatim in configs/estimators/ and configs/closed_loop/.

TABLE I
HYPERPARAMETERS USED ACROSS THE PAPER.

Component	Hyperparameter	Value
Forward model f_θ	state dim	83
	action dim	34
	hidden layers	2×256
	activation	ReLU, LayerNorm
	output	residual (Δx)
	parameters	118,355
Forward training	optimiser	Adam [15]
	learning rate	1×10^{-3}
	weight decay	0
	batch size	256
	epochs	200 (early stop patience 20)
	val split	every 5th source episode
Multi-step loss	rollout horizon H	1, 4, 8
Stage A g_ϕ	input dim	$8 = 3 \text{ ctrl} + 4 \sigma + 1 \text{ delay}/d_{\max}$
	hidden layers	2×32
	output	sigmoid scalar in $[0, 1]$
	parameters	1,409
Stage A training	optimiser	Adam
	learning rate	1×10^{-3}
	weight decay	1×10^{-3}
	epochs	500
Joint-PD controller	K_p, K_d	30, 3
	action scale α	5
	muscle map	$\text{clip}(\alpha \text{ReLU}(M u_{\text{joint}})), M \in \mathbb{R}^{34 \times 20}$
IK solver	method	damped Jacobian pseudoinverse
	max iter / tol	200 / 0.01 m
	damping λ^2	0.1
BC policy	state dim variants	83 (full) / 80 / 77
	hidden layers	2×256
	output	sigmoid 34-dim
	demos / training	30 ep. $\times$ 600 steps; 100 epochs
Eval grid (open-loop)	noise levels	none, low, medium, high, vhigh, xhigh
	delays (steps)	0, 6, 18, 36
	K values	0.0, 0.25, 0.5, 0.75, 1.0
Eval grid (closed-loop)	noise levels (focused)	none, high, xhigh
	delays (focused)	0, 18

APPENDIX B

FORWARD-MODEL ROLLOUT METRICS

Table II reports horizon-wise prediction error for the three forward-model variants used in the paper. Multi-step rollout supervision substantially reduces medium- and long-horizon error while paying a small cost at $h=1$.

APPENDIX C

FULL OPEN-LOOP ORACLE K^* GRIDS

Table III reports the mean oracle K^* per (delay, noise) cell under the $H=8$ multi-step-supervised forward model (used by Sec. IV-E/3.6). The $H=4$ and single-step matrices appear in Fig. 2 of the main text.

TABLE II
FORWARD-MODEL ROLLOUT MSE AND TIP PREDICTION ERROR ON THE HELD-OUT VALIDATION EPISODES. LOWER IS BETTER. ALL THREE MODELS SHARE THE SAME ARCHITECTURE AND TRAINING SET (51,534 TRANSITIONS ACROSS 106 EPISODES); THEY DIFFER ONLY IN THE ROLLOUT HORIZON H .

Model	MSE			Tip err. (m)		
	$h=1$	$h=10$	$h=50$	$h=1$	$h=10$	$h=50$
$H=1$ (single-step)	0.0055	0.0124	0.0524	0.0076	0.0355	0.1381
$H=4$ multi-step	0.0053	0.0084	0.0164	0.0088	0.0247	0.0635
$H=8$ multi-step	0.0055	0.0082	0.0114	0.0098	0.0215	0.0483

TABLE III
ORACLE KALMAN GAIN K^* AS A FUNCTION OF DELAY AND NOISE UNDER THE $H=8$ MULTI-STEP-SUPERVISED FORWARD MODEL (MEAN OVER 3 CONTROLLERS; 72 CONDITIONS TOTAL). 27 OF 72 CELLS REQUIRE $K^* < 1$.

delay	noise					
	none	low	medium	high	vhigh	xhigh
0	1.000	0.833	0.667	0.417	0.333	0.250
6	1.000	1.000	1.000	0.833	0.750	0.500
18	1.000	1.000	1.000	0.917	0.917	0.833
36	1.000	1.000	1.000	1.000	1.000	0.833

APPENDIX D CLOSED-LOOP K -SWEEP BACKING C4

Table IV reports the closed-loop min-tip-to- target error per cell as a function of the (fixed) Kalman gain K , under the $H=8$ forward model and the joint-PD controller (10 episodes per cell). The K -sweep is the basis of the closed-loop oracle K_{cl}^* defined in Sec. IV-F. $K_{cl}^*=0$ in every cell.

Two regularities are worth noting. First, at zero delay the K-curve is sharp around $K=0$: moving from $K=0$ to $K=0.25$ jumps min-tip from 0.49 m to 0.77 m. The Stage A predictor's sigmoid asymptote ($K \approx 0.025$) sits on the wrong side of this jump in delay-0 cells, which is why the residual gap remains (~ 17 cm; see Sec. IV-F). Second, at delay-18 the K-curve is smooth near $K=0$ ($K=0$ and $K=0.25$ within 0.01–0.05 m) so the predictor's small offset is harmless and the same predictor matches the $K=0$ baseline within 1.5 cm.

TABLE IV
CLOSED-LOOP min-TIP ERROR (M) PER CELL ACROSS THE K -SWEEP UNDER THE $H=8$ FORWARD MODEL AND THE JOINT-PD CONTROLLER. EACH CELL AVERAGES 10 EPISODES. BOLD MARKS THE PER-CELL MINIMUM.

noise	delay	$K=0$	$K=0.25$	$K=0.5$	$K=0.75$	$K=1$
none	0	0.493	0.771	0.782	0.774	0.770
none	18	0.497	0.498	0.708	0.746	0.697
high	0	0.493	0.767	0.780	0.771	0.770
high	18	0.497	0.538	0.696	0.746	0.691
xhigh	0	0.493	0.772	0.769	0.766	0.762
xhigh	18	0.497	0.550	0.713	0.749	0.700

APPENDIX E TRANSFER TEST ON REACHRANDOM

Setup. We re-run a slim version of the pipeline on myoArmReachRandom-v0 (same 34-muscle arm,

same 600-step episodes, random per-episode target instead of a fixed one). Targets come from the same runs/targets/train.npz file (200 positions) so that targets are paired across estimators and forward models; initial joint pose is deterministic ($q=0$) for both variants. Transition data: 49,740 transitions across 106 episodes (random / lowamp / hold controllers), comparable to the 51,534- transition ReachFixed dataset. The marginal C2 cell-level effect (≈ 8.6 cm at one cell with learned $K \approx 0.99$ and paired- t $p=0.21$) is task-dependent and *not* treated as a transfer target; the test asks whether the structural phenomena C1 / C3 / C4 persist under variable-target reaching. Pre-committed pass criteria, fixed before reading the numbers:

- *C1 replicate*: open-loop oracle K_{ol}^* varies with noise / delay, with long-delay cells biasing toward $K=1$.
- *C3 replicate*: multi-step supervision reduces $h=50$ forward-model error and increases the share of closed-loop cells where $K=0$ or prediction-heavy estimators beat $K=1$.
- *C4 replicate*: a closed-loop oracle K_{cl}^* trained predictor moves toward the $K=0$ / K_{cl}^* side of the K-curve compared with an open-loop oracle predictor.

Forward-model rollout error (Table V) mirrors the ReachFixed pattern: $H=8$ multi-step rollout supervision reduces $h=50$ tip prediction error by 48% versus $H=1$ (0.110 $\rightarrow$ 0.057 m).

TABLE V
REACHRANDOM FORWARD-MODEL ROLLOUT MSE AND TIP PREDICTION ERROR ON HELD-OUT VALIDATION EPISODES. SAME NETWORK AND TRAINING RECIPE AS THE MAIN TEXT; LOWER IS BETTER.

Model	MSE			Tip err. (m)		
	$h=1$	$h=10$	$h=50$	$h=1$	$h=10$	$h=50$
$H=1$ (single-step)	0.0045	0.0103	0.0318	0.0066	0.0291	0.1098
$H=8$ multi-step	0.0046	0.0072	0.0096	0.0093	0.0229	0.0573

Closed-loop reaching, single-step model. Table VI reports min-tip error at the 6-cell focused grid for $K=0$ and $K=1$ under the joint-PD controller. Two observations: (i) $K=0$ is constant across noise/delay because it ignores observations, and (ii) on ReachRandom $K=0$ already beats $K=1$ in *every* cell, by 7–27 cm. This contrasts with ReachFixed where the $H=1$ closed-loop optimum was nearer $K=1$ at delay-18 cells; the gap is therefore task-dependent at the single-step baseline.

TABLE VI
REACHRANDOM $H=1$ CLOSED-LOOP min-TIP ERROR (M, MEAN OVER 10 EPISODES) UNDER THE JOINT-PD CONTROLLER. $K=0$ BEATS $K=1$ IN EVERY CELL BY 7–27 CM.

noise	delay	$K=0$	$K=1$	$K=1-K=0$
none	0	0.499	0.770	+27.2 cm
none	18	0.498	0.578	+8.0 cm
high	0	0.499	0.770	+27.2 cm
high	18	0.498	0.567	+6.8 cm
xhigh	0	0.499	0.762	+26.3 cm
xhigh	18	0.498	0.593	+9.4 cm

Closed-loop K-curve, $H=8$ multi-step model. Table VII reports the full $K \in \{0, 0.25, 0.5, 0.75, 1\}$ sweep at $H=8$. The closed-loop oracle K_{cl}^* is the per-cell minimiser. Unlike ReachFixed, where $K_{cl}^*=0$ in every cell of the focused grid, on ReachRandom K_{cl}^* splits across three values: $K=0$ at all three $d=0$ cells (matching ReachFixed), $K=0.25$ at delay-18 in the low/medium-noise cells, and $K=1$ at the (xhigh, $d=18$) cell.

TABLE VII

REACHRANDOM $H=8$ CLOSED-LOOP min-TIP K-SWEEP (MEAN OVER 10 EPISODES). BOLD MARKS THE PER-CELL K_{cl}^* .

noise	delay	$K=0$	$K=0.25$	$K=0.5$	$K=0.75$	$K=1$
none	0	0.670	0.762	0.785	0.780	0.770
none	18	0.642	0.625	0.706	0.644	0.647
high	0	0.670	0.764	0.785	0.775	0.770
high	18	0.642	0.631	0.688	0.651	0.639
xhigh	0	0.670	0.758	0.765	0.761	0.762
xhigh	18	0.642	0.662	0.691	0.650	0.627

Open-loop oracle structure on ReachRandom. We re-ran the full 3 controllers $\times$ 6 noise $\times$ 4 delay $\times$ 5 K stress eval (using 2 episodes per controller, matching the ReachFixed scale) for both forward models. $K_{ol}^* < 1$ holds in 18 / 72 cells under $H=1$ (exactly matching ReachFixed at $H=1$) and 25 / 72 cells under $H=8$ (vs. 27 / 72 on ReachFixed); blending is again confined to delay-0 cells under $H=1$ and expands to delay ≥ 6 under $H=8$ (Table VIII). C1 therefore replicates structurally on ReachRandom.

TABLE VIII

REACHRANDOM K_{ol}^* (MEAN OVER 3 CONTROLLERS) PER (NOISE, DELAY) CELL, UNDER $H=1$ (TOP) AND $H=8$ (BOTTOM). DASHES MARK $K_{ol}^*=1.0$ (TRUST THE OBSERVATION). THE NON-TRIVIAL $K_{ol}^* < 1$ REGIME IS CONFINED TO DELAY-0 UNDER $H=1$ AND SPREADS TO DELAY ≥ 6 UNDER $H=8$, MIRRORING REACHFIXED.

$H=1$ single-step forward model				
noise \ delay	0	6	18	36
none	—	—	—	—
low	0.83	—	—	—
medium	0.67	—	—	—
high	0.50	—	—	—
vhigh	0.33	0.92	—	—
xhigh	0.25	0.67	—	—
$H=8$ multi-step forward model				
noise \ delay	0	6	18	36
none	—	—	—	—
low	0.83	—	—	—
medium	0.67	—	—	—
high	0.42	0.92	—	—
vhigh	0.33	0.67	0.92	—
xhigh	0.25	0.50	0.75	0.75

Learned predictors trained on K_{ol}^* vs K_{cl}^* . We train the same $8 \rightarrow 32 \rightarrow 32 \rightarrow 1$ Stage A predictor on (i) the K_{ol}^* table from the $H=8$ ReachRandom stress eval, and (ii) the per-cell K_{cl}^* labels from Table VII (extended to the 72-cell schema by replicating the joint-PD label across the auxiliary controllers and the unswept delays/noises). Table IX reports the K each predictor outputs at the focused grid and the resulting closed-loop min-

tip. The K_{ol}^* predictor recovers values near $K=1$ at delay-18 cells (its training labels were $K_{ol}^*=1$ there), matching the $K=1$ baseline within 1 cm. The K_{cl}^* predictor recovers small K (0.06–0.31) at delay-0 cells, beating the K_{ol}^* predictor by 1.3–12.6 cm there. At delay-18 the K_{cl}^* predictor's outputs drift toward intermediate values ($K=0.24$ –0.95) and the resulting closed-loop reach is comparable to $K=1$ (within ± 8 cm). The oracle-redefinition transfer therefore holds clearly at delay-0 but is muted at delay-18 because the closed-loop K-curve there is relatively flat (Table VII) and the sigmoid output cannot cleanly land on the mixed K_{cl}^* (0/0.25/1) values across the three noise rows.

TABLE IX

REACHRANDOM $H=8$ CLOSED-LOOP COMPARISON OF FOUR ESTIMATORS ON THE FOCUSED GRID. COLUMNS: $K=0$, $K=1$, *learned-ol* (STAGE A TRAINED ON K_{ol}^*), *learned-cl* (STAGE A TRAINED ON K_{cl}^*); FOR EACH LEARNED PREDICTOR WE REPORT BOTH ITS MEAN OUTPUT $\bar{K}$ AND THE CLOSED-LOOP min-TIP (M, MEAN OVER 10 EPISODES).

noise	delay	$K=0$	$K=1$	<i>learned-ol</i>		<i>learned-cl</i>	
				$\bar{K}$	m	$\bar{K}$	m
none	0	0.670	0.770	0.80	0.781	0.06	0.655
none	18	0.642	0.647	0.98	0.638	0.24	0.673
high	0	0.670	0.770	0.71	0.777	0.09	0.690
high	18	0.642	0.639	0.96	0.644	0.47	0.721
xhigh	0	0.670	0.762	0.36	0.783	0.31	0.770
xhigh	18	0.642	0.627	0.88	0.654	0.95	0.642

Assessment against the pre-committed criteria.

- *C1 (full replication)*: open-loop oracle K_{ol}^* varies with noise and delay in the same qualitative shape as ReachFixed, with the $K_{ol}^* < 1$ fraction growing from 18 / 72 ($H=1$) to 25 / 72 ($H=8$).
- *C3 (partial replication)*: $H=8$ rollout supervision reduces forward-model error by 48 % (replicates) and $K=0$ is optimal at all three delay-0 cells (replicates). The ReachFixed-style *regime shift* from $K=1$ -winning at $H=1$ to $K=0$ -winning at $H=8$ does *not* apply because $K=0$ was already winning at $H=1$; the variable-target variant is structurally more favourable to prediction-only across forward-model strengths.
- *C4 (partial replication, made richer)*: the closed-loop oracle K_{cl}^* on ReachRandom is *not* uniformly $K=0$ (3 cells pick $K=0$, 2 cells pick $K=0.25$, 1 cell picks $K=1$), confirming K_{cl}^* is a non-degenerate task-and-cell-dependent function. A Stage A predictor retrained on these labels matches the $K=0$ baseline within 1.3 cm at every delay-0 cell and beats the K_{ol}^* -trained predictor by 1–13 cm there. At delay-18, where K_{cl}^* takes three different values across the three noise rows, the predictor's sigmoid output cannot cleanly land on each label, and the closed-loop benefit is muted.

Take-away. The ReachRandom transfer test does not falsify any C1/C3/C4 claim; it refines them. C1 replicates fully. C3's regime-shift framing is task-dependent, but the underlying fact (multi-step supervision makes prediction-only competitive in closed loop) replicates. C4's oracle-redefinition argument transfers cleanly to delay-0 cells; at delay-18 the sigmoid-output predictor cannot fully exploit the cell-dependent K_{cl}^* ,

which is the same parameterisation limitation flagged in Sec. [IV-F](#).

APPENDIX F

SOFTWARE, HARDWARE, AND REPRODUCIBILITY

All experiments run on a single workstation (no GPU is required; the forward model and Stage A predictor are small enough that CPU training takes only minutes). Software stack: Python 3.11, PyTorch (CPU), MuJoCo 2.3, MyoSuite 2.12.2, Gymnasium 0.29. Reproduction scripts and exact dataset paths are recorded in the source repository's `docs/02_InitialImplementationPlan.md` and `docs/03_PaperOutline.md`.